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Chapter 8: Energy and Life in Plants

Photosynthesis is the biological process by which green plants, algae, and cyanobacteria convert solar light energy into chemical energy stored in glucose. This critical reaction takes place within specialized organelles called chloroplasts, which contain the light-absorbing pigment chlorophyll.

The balanced chemical equation for photosynthesis is:
6CO2 + 6H2O + Sunlight -> C6H12O6 + 6O2

Carbon dioxide enters the leaves through microscopic pores called stomata, while water is drawn up from the soil through root xylem vessels. During the light-dependent reactions, sunlight splits water molecules (photolysis), releasing oxygen as a byproduct and charging ATP and NADPH molecules.

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Stage 2: The Calvin Cycle (Light-Independent Reactions)
The Calvin cycle occurs in the stroma, the fluid-filled interior of the chloroplast. Using the chemical energy carried by ATP and NADPH, carbon dioxide from the atmosphere is fixed into carbohydrate building blocks like G3P.

Key limiting factors affecting the rate of photosynthesis include:
1. Light intensity: Increases rate up to a saturation threshold.
2. Carbon dioxide concentration: Higher availability accelerates Calvin cycle fixing.
3. Temperature: Moderate warmth promotes enzyme action; above 40C, RuBisCO denatures.